SCST1223: PROBABILITY & STATISTICAL DATA ANALYSIS
2024/2025 – SEMESTER 2

PROJECT 1

The aim of Project 1 is to give you experience in data collection and analysis. The data analysis is only based on **descriptive statistics.** For example, find the frequency distribution, displaying the categorical data using bar chart, displaying the numerical data using stem and leaf plot, compute the parametric values (mean, median, mod) and etc.

You need to conduct this project in a **group of 3 or 4 students**. Discuss with your group members to identify any subject of interest of the study. The examples of topics that could be studied as follows:

1) The use of the Internet in learning during pandemic Covid-19.
2) The use of social media to improve sales.

In this project you are required to collect a sample data. Your sample size must be at least 60 data. The data collection must use questionnaire. Include the questionnaire in the report. To analyze the data, **you must use R programming**. All the analysis must be compiled and written as a report. The organization of your report is as follows:

1) **Introduction or background**
   Explain the purpose of the study in your own words, why you were interested in the question and what you expected to see from the data.
2) **Data Collection**
   Data description including parameters or variable to be measured.
3) **Data Analysis**
   a. Categorical data:
      - **Graphs**: Pie chart or bar chart (with percent not counts)
      - **Numerical**: Proportion and sample size in each category
   b. Quantitative data:
      - **Graphs**: E.g., histograms or dot plots.
      - **Measures of Central Tendency**: E.g., mean, media, boxplots

   Interpret and discuss what you can learn from the pictures (e.g., compare shape…) and make sure you report the sample sizes for the survey as well.
4) **Conclusion**
   Write a one-paragraph summary on what you learned about the population you surveyed. Explain what did you learn mathematically and/or technologically from conducting this study? Any final thoughts.

Report must be submitted through elearning.utm.my on/before **Week 6 (21 – 25 April 2025; 5 PM)**.